IBM Hackathon Submission

Vyuit / Doori
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Written problem and solution statement

Taking possession of a new apartment from a contractor can be a demanding process with financial implications. People without construction knowledge may struggle to know what to inspect, understand professional terminology and document their observations clearly.

I built Vyuit to help users approach this process through room-by-room guidance, photographs, written findings and a report. The app draws on my architectural experience. It is intended to support informed observation and documentation, not replace a professional inspection or certify that an apartment is free of defects.

Initial testing and conversations with potential users revealed uncertainty about inspecting independently and, among some, a preference for hiring a professional inspector. One hypothesis I want to test is whether conversational guidance can help people who find written checklists difficult to follow.

During the hackathon, I began developing Doori, a Hebrew-speaking guide agent using IBM watsonx Orchestrate and Build with Bob. The existing app was developed in Lovable before the hackathon. I created a separate development branch and worked with Bob on the agent configuration, conversation instructions, knowledge files and initial checks. I provided the professional content, inspection sequence and product decisions.

The intended interaction combines guidance, explanation and documentation within the relevant room. For example, in Bathroom 1, Doori asks one focused question, waits for the user's response and explains unfamiliar terminology when needed. If the user reports a finding, Doori helps formulate a factual description. The user remains responsible for reviewing the wording, attaching a photograph and saving the finding in the app.

The prototype includes a cloud-hosted agent and knowledge base. Conversations, knowledge retrieval and the embedded chat widget were tested. However, adherence to the inspection sequence remains inconsistent, and not all locally revised instructions have been imported and verified in the cloud. Tools for tracking answered items and controlling progression were planned but have not been implemented. The agent does not write findings to the app's database.

A separate interactive HTML demo illustrates the intended end-to-end experience, including conversation, editing a finding, simulated saving and a summary. Its conversation is scripted, and its saving is simulated. It is not connected to the live agent. The submission therefore distinguishes the tested prototype from the experience envisioned for further development.

Initial visual design work presents Doori as a floating door character. Planned movement of its body and handle could communicate waiting, explanation and progress. Animation and its effect on usability have not yet been implemented or tested.

Existing app (pilot access): https://vyuit.app